Diafiltration Process Model: Mass Balance & Thermodynamics (v10)

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Contents

1 Roadmap and diagram associations

We proceed: **Bulk (retentate) → CP film → Membrane → Permeate path**. This draft has a focus on mass balances, thermodynamics, and transport mechanisms.

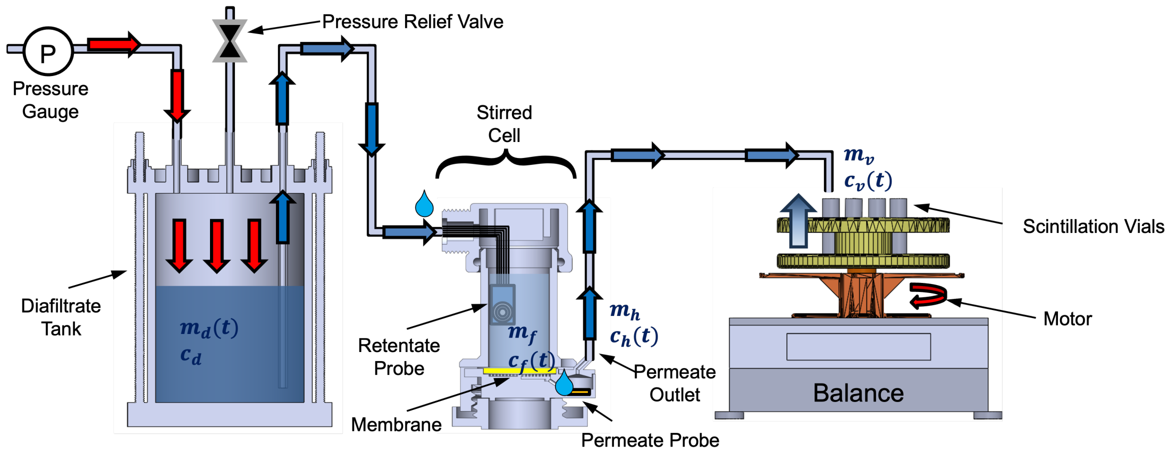

Figure 1: Process variables of importance to the diafiltration system.

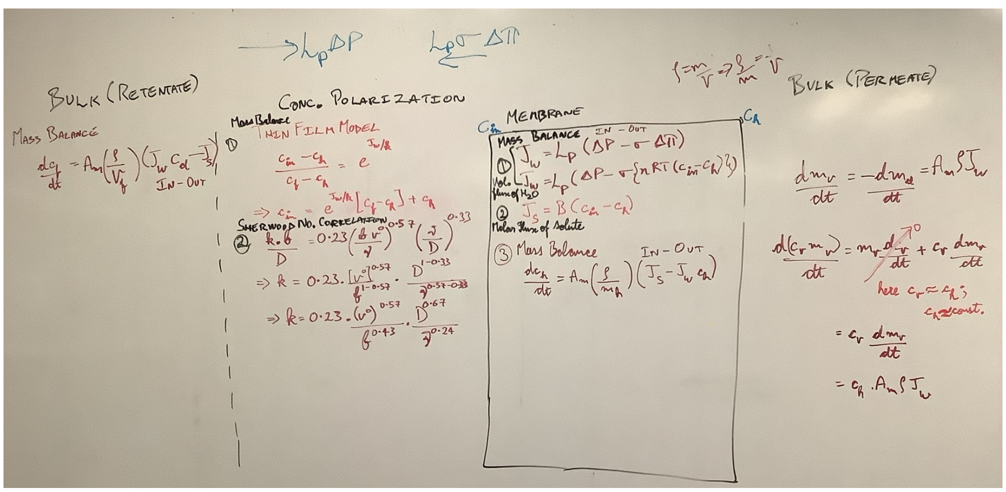

Figure 2: Near-membrane profile: bulk → CP film → membrane interfaces.

Assumptions and Deviations from Reference Models

- **Geometry**: Planar membrane; only the cross-membrane direction is modeled.
- **Hydrodynamics**: Retentate is well-mixed; a steady CP film is represented via a film coefficient k_i. Permeate holdup/vials are treated as well-mixed averaging elements.
- **Thermodynamics**: Electrochemical potential continuity, Donnan partition, steric hindrance, and electroneutrality constraints as in Supporting Information (Sxx). Activities may be introduced for high ionic strength.
- **Transport**: Water flux via Kedem–Katchalsky; solute transport via an ENP-based reduction to a flux with interface partitions $H_{f,i}$, $H_{p,i}$.
- **Scope**: Only mathematical modeling and derivations; experimental/control/estimation protocols are out of scope here.

2 Bulk (retentate) → CP

2.1 Startup modes and total mass balance

Let m_f be the retentate mass, ρ density, A_m area, J_w water flux (to permeate), and $\dot{m}_d$ the diafiltrate inflow.

$$\text{Lag:} \quad \frac{dm_f}{dt} = -\rho A_m J_w, \quad \dot{m}_d = 0, \tag{1}$$

$$\text{Overflow:} \quad \frac{dm_f}{dt} = \dot{V}_o - \rho A_m J_w, \quad \dot{m}_d = \dot{V}_o, \tag{2}$$

$$\text{Post-startup:} \quad \frac{dm_f}{dt} = \dot{V} - \rho A_m J_w, \quad \dot{m}_d = \dot{V}. \tag{3}$$

[cf. §5.1 From continuity to startup/retentate balances]

2.2 Species balances (vector form)

For species i, retentate concentration $c_{f,i}(t)$, diafiltrate composition $c_{d,i}$, membrane solute flux $J_{s,i}$, and permeate holdup concentration $c_{h,i}$:

$$\frac{d}{dt}(m_f c_f) = \dot{m}_d\, c_d - \rho A_m J_w c_h - A_m J_s, \tag{4}$$

$$\Rightarrow \quad \frac{dc_f}{dt} = \frac{1}{m_f}\left[\dot{m}_d(c_d - c_f) - A_m J_s - \rho A_m J_w (c_h - c_f)\right]. \tag{5}$$

[cf. §5.1]

2.3 CP-film closure

Define feed-side interface concentrations $c_{\mathrm{in},f,i}$ and film coefficient k_i (species-dependent allowed). With reflection σ_i in the film (optional):

$$\frac{c_{\mathrm{in},f,i} - c_{h,i}}{c_{f,i} - c_{h,i}} = \exp\left[\frac{J_w(1-\sigma_i)}{k_i}\right], \tag{6}$$

[cf. §5.2 CP-film exponential from 1-D convection–diffusion]

The Sherwood correlation reads

$$\mathrm{Sh}_i = \frac{k_i b}{D_{s,i}} = a\left(\frac{v_0 b}{\nu}\right)^{b_1}\left(\frac{b}{2D_{s,i}}\right)^{b_2}, \tag{7}$$

[cf. §5.2; van den Berg (1989); Gekas & Hallström (1987)]

3 Concentration Polarization (CP) → Membrane (transport and partition)

3.1 Water flux and osmotic models

$$J_w = L_p(\Delta P - \sigma_w \Delta\pi), \qquad \Delta\pi = \sum_i n_i RT(c_{\mathrm{in},f,i} - c_{\mathrm{in},p,i}), \tag{8}$$

[cf. §5.3 Kedem–Katchalsky for water flux]

3.2 Charged–membrane solute transport

Mechanistic convection–diffusion solution. For each i across membrane thickness L with effective diffusivity $D_{m,i}$ and partition coefficients $H_{f,i}, H_{p,i}$:

$$J_{s,i} = J_w \big[H_{f,i} c_{\text{in},f,i} e^{\text{Pe}_i} - H_{p,i} c_{\text{in},p,i}\big] / (e^{\text{Pe}_i} - 1), \qquad \text{Pe}_i = \frac{J_w L}{D_{m,i}}, \tag{9}$$

[cf. §5.4 Solute flux $J_{s,i}$ across a slab; §5.6 Fixed–charge ENP reduction]

In the diffusive limit ($\text{Pe}_i \ll 1$), $J_{s,i} \approx (D_{m,i} H_i / L)(c_{\text{in},f,i} - c_{\text{in},p,i})$ with $H_i \simeq H_{f,i} \simeq H_{p,i}$. [cf. §5.4]

Donnan and steric partition (interface) For fixed charge density X_f (mol m^{-3}-pore) and solute valence z_i (*feed interface* $z=0$, *permeate interface* $z=L$):

$$c_{m,\text{in},\ell,i} = \Phi_i \, c_{\text{in},\ell,i} \exp\left(-\frac{z_i F \Delta\psi_{D,\ell}}{RT}\right), \qquad \ell \in \{f, p\}, \tag{10}$$

with steric factor $\Phi_i = (1 - \lambda_i)^2$ and $\lambda_i = a_i / r_p$. Electroneutrality holds within each interfacial control volume:

$$\sum_i z_i c_{m,\text{in},\ell,i} + X_f = 0.$$

[cf. §5.5 Donnan + steric partition from electrochemical potentials; §5.6 Fixed–charge ENP reduction]

Intrinsic and observed sieving. Using the zero–current constraint (no external current) from § 5.6.1, the effective charge–coupled transport through the membrane may be expressed as

$$S_i^* = \Phi_i \exp\left(-\frac{z_i F \Delta\psi_{D,f}}{RT}\right) \frac{P_i^{\text{ENP}}}{P_i^{\text{ENP}} + J_w}, \tag{11}$$

where P_i^{ENP} is the effective ENP permeability obtained by integrating the reduced fluxes across L [cf. §5.6.1 No–current constraint and intrinsic charged sieving].

Including concentration polarization via the exponential film correction (Eq. 6) yields the observed sieving:

$$S_i^{\text{obs}} = S_i^* \exp\left(\frac{J_w}{k_i}\right), \tag{12}$$

which links interface phenomena, membrane electrostatics, and external mass transfer into a unified predictive expression for charged–solute rejection.

Empirical permeability family. For semi–empirical parameterization,

$$J_{s,i} = B_i(J_w, c_{\text{in},f,i})\big(c_{m,\text{in},f,i} - c_{m,\text{in},p,i}\big), \qquad B_i = J_w(\beta_{0,i} + \beta_{1,i} c_{\text{in},f,i}), \tag{13}$$

[cf. §5.6 and §5.6.1] offering a convenient regression form that recovers Eq. 12 when B_i is related to P_i^{ENP} and $H_{f,i}$.

4 Membrane → Permeate path

4.1 Holdup, tubing, and vial averaging

Let (m_h, c_h) be the holdup state under the membrane, tubing mass m_{tb}, and vial state (m_v, c_v).

$$\text{Holdup fill } (t_0 \le t < t_H): \frac{dm_h}{dt} = \rho A_m J_w, \quad \frac{d(m_h c_h)}{dt} = A_m J_s, \tag{14}$$

$$\text{Steady holdup } (t_H \le t < t_V): \frac{dm_h}{dt} \approx 0, \quad \frac{d(m_h c_h)}{dt} \approx A_m J_s - q_{\mathrm{out}}, \tag{15}$$

$$\text{Vial fill } (t \ge t_V): \frac{dm_v}{dt} = \rho A_m J_w, \quad \frac{d(m_v c_v)}{dt} = c_h \frac{dm_v}{dt}. \tag{16}$$

Averaging over a collection window $[t_k, t_{k+1}]$ gives

$$c_v^{(k)} = \frac{1}{\Delta m_v^{(k)}} \int_{t_k}^{t_{k+1}} c_h(t)\, \rho A_m J_w(t)\, dt, \quad \Delta m_v^{(k)} = m_v(t_{k+1}) - m_v(t_k).$$

Time alignment uses $\tau_{\mathrm{tb}} = m_{\mathrm{tb}}/(\rho A_m \bar{J}_w)$. [cf. §5.7 Permeate holdup and vial averaging]

References